

Advanced (Habanero)**Q1.** What is the standard form of a quadratic function?

- (A) $ax^2 + bx + c$
 (B) $ax + bx^2 + c$
 (C) $ax^2 - bx + c$
 (D) $ax^3 + bx^2 + c$
 (E) $ax^2 + bx - c$

quadratic function $x^2 + 4x + 4$?

- (A) $x = -2$
 (B) $x = -1$
 (C) $x = 0$
 (D) $x = 1$
 (E) $x = 2$

Q2. Which of the following is a quadratic function in standard form?

- (A) $x^2 + 3x - 2$
 (B) $x^3 - 2x^2 + x$
 (C) $x^2 - 4x + 4$
 (D) $x^4 + 2x^2 + 1$
 (E) $x^2 + 2x - 3$

Q6. Which of the following is a key feature of the quadratic function $x^2 - 4x + 4$?

- (A) A maximum value
 (B) A minimum value
 (C) A zero at $x = 2$
 (D) A vertical asymptote
 (E) A horizontal asymptote

Q3. What is the value of a in the quadratic function $2x^2 + 3x - 1$?

- (A) 1
 (B) 2
 (C) 3
 (D) -1
 (E) -2

Q7. What is the value of the quadratic function $x^2 + 2x - 3$ when $x = -1$?

- (A) 0
 (B) 1
 (C) 2
 (D) 3
 (E) 4

Q4. Which of the following describes the graph of the quadratic function $x^2 + 2x + 1$?

- (A) A parabola that opens up
 (B) A parabola that opens down
 (C) A horizontal line
 (D) A vertical line
 (E) A circle

Q8. Which of the following describes the quadratic function $-x^2 + 2x - 1$?

- (A) A parabola that opens up
 (B) A parabola that opens down
 (C) A horizontal line
 (D) A vertical line
 (E) A circle

Q5. What is the equation of the axis of symmetry of the**Open Ended Questions (FRQ)****Q9.** Graph the quadratic function $x^2 + 2x - 3$ and identify its key features.**Q10.** Write the quadratic function in standard form that has its vertex at $(-2, 1)$ and passes through the point $(0, -3)$.**Chef's Tips**

- Tip 1: Use graph paper for questions 9 and 10.
- Tip 2: Check for any calculations or graphs that are not clearly labeled.
- Tip 3: Make sure to read each question carefully and follow the instructions.